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Masterarbeit

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Abstract

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.

Allgemeine Merkmale

- **Objektivität:** Es soll sich jeder persönlichen Wertung enthalten.
- **Kürze:** Es soll so kurz wie möglich sein.
- **Verständlichkeit:** Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- **Vollständigkeit:** Alle wesentlichen Sachverhalte sollen enthalten sein.
- **Genauigkeit:** Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

1 Einleitung

2 Literature Review

2.1 Maintenance Optimization in Infrastructure Systems

Maintenance optimization in infrastructure systems addresses the challenge of planning, scheduling, and executing maintenance activities to ensure system availability, reliability, and cost-efficiency over the lifecycle of technical assets. Unlike manufacturing or logistics contexts, infrastructure maintenance often involves geographically distributed assets, limited maintenance resources, and significant consequences of system failures (de Jonge & Scarf, 2020; Dekker, 1996).

This chapter provides the conceptual foundation for maintenance optimization. Section 2.1.1 defines maintenance optimization and its objectives. Section 2.1.2 reviews maintenance strategies: preventive, reactive, and condition-based. Section 2.1.3 discusses uncertainty, availability, and cost as central decision variables. Section 2.1.4 provides the transition to mobile maintenance and service interventions (de Jonge & Scarf, 2020; Alaswad & Xiang, 2017).

2.1.1 Concept and Objectives of Maintenance Optimization

Maintenance optimization refers to the systematic determination of when and what to maintain in order to achieve specific objectives such as minimizing costs, maximizing availability, or balancing both (de Jonge & Scarf, 2020). The fundamental trade-off lies between the cost of maintenance activities and the cost of system failures or downtime (Dekker, 1996).

Early work by Dekker (1996) identified six barriers preventing the application of maintenance optimization models in practice: models are difficult to understand, many papers are written only for mathematical purposes, maintenance activities consist of many different aspects, optimization is not always necessary, models often focus on the wrong type of maintenance, and there is a gap between theory and operational reality. Despite these challenges, maintenance optimization remains strategically important because continuous developments of technical systems and increasing reliance on equipment have resulted in growing importance of effective maintenance activities (de Jonge & Scarf, 2020).

In the context of charging infrastructure, maintenance optimization must address not only individual station failures but also fleet-level dispatching, resource allocation, and timing decisions under uncertainty (de Jonge & Scarf, 2020). The objective is typically to minimize total expected costs (maintenance plus failure costs) while maintaining a required level of system availability (Alaswad & Xiang, 2017; Dekker, 1996).

2.1.2 Maintenance Strategies: Preventive, Reactive, Condition-Based

Maintenance strategies can be broadly classified into three categories: reactive (corrective) maintenance, preventive (planned) maintenance, and condition-based (predictive) maintenance (Sarja & Halilov, 2023; de Jonge & Scarf, 2020).

Reactive maintenance (also called corrective or run-to-failure maintenance) is performed

only after a failure occurs (Sarja & Halilov, 2023). This strategy is suitable for non-critical assets where downtime has minimal impact on operations. However, for critical infrastructure such as charging stations, reactive maintenance can lead to significant customer dissatisfaction, safety risks, and higher repair costs due to unanticipated failures (de Jonge & Scarf, 2020).

Preventive maintenance is performed according to a fixed schedule or usage-based interval, regardless of the actual condition of the asset (Sarja & Halilov, 2023; Alaswad & Xiang, 2017). Preventive maintenance aims to prevent failures before they occur by replacing or servicing components at predetermined times. This approach is well-established and widely applied, for example in vehicle maintenance or periodic inspections of technical systems (Dekker, 1996). However, preventive maintenance can lead to over-maintenance (replacing components that are still functional) or under-maintenance (failures occurring before the scheduled interval) (de Jonge & Scarf, 2020).

Condition-based maintenance (CBM) performs maintenance based on the actual condition of the asset, which is monitored through sensors, inspections, or performance data (Alaswad & Xiang, 2017; Sarja & Halilov, 2023). CBM enables maintenance to be performed exactly when needed, reducing unnecessary interventions and focusing resources on assets that actually require attention. In the literature, CBM is often extended to predictive maintenance, which uses data analytics, machine learning, or digital twins to predict when maintenance will be needed (Sarja & Halilov, 2023).

The review by Alaswad and Xiang (2017) on condition-based maintenance optimization models for stochastically deteriorating systems shows that CBM is particularly suitable for systems where condition monitoring is feasible and failure consequences are significant. For charging infrastructure, this includes monitoring of charging power, error rates, temperature, or communication status (de Jonge & Scarf, 2020).

In practice, many organizations use a hybrid strategy, combining preventive maintenance for critical components with reactive maintenance for non-critical ones and condition-based maintenance for assets where monitoring is available (Sarja & Halilov, 2023; de Jonge & Scarf, 2020). For this thesis, the hybrid perspective is relevant because maintenance of charging infrastructure often involves both planned inspections and unplanned failure responses.

2.1.3

Maintenance decisions are inherently made under uncertainty. Key uncertain parameters include failure times, service times, travel times, availability of spare parts, and arrival of new maintenance requests (de Jonge & Scarf, 2020; Dekker, 1996). This uncertainty affects three central decision variables:

Availability measures the proportion of time that a system is operational and able to fulfill its function (Sarja & Halilov, 2023). For charging infrastructure, availability is a key performance indicator because failed stations directly reduce customer access and revenue. Predictive maintenance can increase availability by 5–15 % according to McKinsey estimates, while reducing maintenance costs by 18–25 % (Sarja & Halilov, 2023).

Cost includes maintenance costs (labor, materials, travel), failure costs (downtime, customer dissatisfaction, penalties), and inventory costs (spare parts, tools) (de Jonge & Scarf, 2020; Dekker, 1996). The optimization objective is typically to minimize total expected cost over a planning horizon, subject to availability constraints (Alaswad & Xiang, 2017).

Uncertainty affects both the timing and the impact of maintenance decisions. For example, a failure may occur earlier or later than expected, service may take longer than planned, or travel to a site may be delayed by traffic (de Jonge & Scarf, 2020). Decision-making under uncertainty follows a sequential process in which uncertainty realization (observation of the true values of previously uncertain parameters) and subsequent reactive decision-making alternate as time proceeds (de Jonge & Scarf, 2020).

In maintenance optimization, uncertainty is often modeled using stochastic deterioration processes, probability distributions for failure times, or scenario-based approaches (Alaswad & Xiang, 2017; de Jonge & Scarf, 2020). The choice of uncertainty model depends on the amount of available data, the complexity of the system, and the computational requirements of the optimization approach.

2.1.4 Transition to Mobile Maintenance and Service Interventions

The maintenance problems discussed in Sections 2.1.1–2.1.3 typically focus on when and what to maintain at individual assets. However, in many practical settings, maintenance is performed by mobile technicians who must travel between geographically distributed sites, which introduces routing and scheduling decisions (de Jonge & Scarf, 2020).

For charging infrastructure maintenance, this means that the optimization problem is not only about determining Maintenance intervals or failure thresholds but also about dispatching technicians, planning routes, and allocating limited resources (vehicles, spare parts, time) across multiple stations (de Jonge & Scarf, 2020; Alaswad & Xiang, 2017).

This transition from single-asset maintenance optimization to mobile maintenance service planning introduces additional complexity:

- Spatial dependency: Maintenance decisions for one station affect the availability of resources for other stations (de Jonge & Scarf, 2020).
- Temporal dependency: The order in which stations are visited affects total travel time and service availability (Alaswad & Xiang, 2017).
- Uncertainty coupling: Failures, travel times, and service times are often stochastic and interdependent (de Jonge & Scarf, 2020). The literature on maintenance optimization has traditionally focused on single-asset or multi-asset models without explicit routing (de Jonge & Scarf, 2020; Dekker, 1996). More recent work begins to integrate maintenance planning with vehicle routing, especially in the context of dynamic and stochastic vehicle routing problems (SDVRP) (de Jonge & Scarf, 2020). This integration is precisely the focus of the following chapters 2.2 and 2.3, which cover stochastic routing and approximate dynamic programming for dynamic service planning.

For this thesis, the key insight from Chapter 2.1 is that maintenance optimization provides the decision objectives (cost, availability, uncertainty handling), while Chapters 2.2 and 2.3 provide the methodological framework (SDVRP, ADP) for implementing these objectives in a mobile, dynamic, and stochastic service environment (de Jonge & Scarf, 2020; Alaswad

& Xiang, 2017).

3 Zitieren und Referenzieren

Beiträge in Fachzeitschriften wie Clemen (1989) oder Konferenzartikel wie He u. a. (2017) werden auf diese Weise im Text zitiert. In anderen Fällen möchte man aber in Klammern zitieren (Clemen, 1989), auch mit mehreren Autoren (Clemen, 1989; Baumol und Wolfe, 1958; He u. a., 2017).

Bei Monographien muss eine Seitenzahl mit angegeben werden (Chollet, 2018, S. 28).

So wird eine Webquelle zitiert: RStudio (2017). Es kann bei kurzen Informationen im Internet aber auch reichen die Adresse¹ als Fußnote einzubetten.

So werden andere Teile der Arbeit referenziert: Kapitel 1, Gleichung 1 zeigt...

So verweisen wir auf eine Fußnote ².

4 Abbildungen

Abbildungen erfordern das package *graphicx*. Idealerweise verwendet man Vektorgrafiken oder hochaufgelöste Bitmaps. Eine gute Variante ist das Verwenden von PDFs.



Abbildung 1: Siegel der Universität

5 Tabellen

Die Tabular-Umgebung gibt die Anzahl Spalten an, deren Orientierung, Breite und evtl. Zwischenlinien.

¹<https://shiny.rstudio.com/tutorial/written-tutorial/lesson1/>

²dies ist eine Fußnote

Tabelle 1: Meine Tabelle

col1	col2	col3
Multiple row	cell2	cell3
	cell5	cell6
	cell8	cell9

6 Formeln

$$\sum_{i=1}^N x_i \tag{1}$$

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A Anhang A

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.

Ich erkläre weiterhin, dass die vorliegende Arbeit in gleicher oder ähnlicher Form noch nicht im Rahmen eines anderen Prüfungsverfahrens eingereicht wurde.

Würzburg, den 17. Juni 2026

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